

The $3BF$ state sum as a TQFT

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A SKETCH OF THE TALK

- Preliminaries
 - ▶ 3-group and 3-gauge theory
 - ↪ based on *R. Picken and J. Faria Martins*, [arXiv:0907.2566](#).
 - ▶ $3BF$ action
 - ↪ Models with relevant dynamics based on *A. Miković and M. Vojinović*, [arXiv: 1110.4694](#), and *TR and M. Vojinović*, [arXiv:1904.07566](#).
 - ▶ Quantization of the topological $3BF$ theory
 - ↪ the state sum Z is an example of Porter's TQFT for $d = 4$ and $n = 3$ *T. Porter* (1998), based on *TR and M. Vojinović*, [arXiv: 2201.02572](#).
 - ↪ The construction of the $3BF$ state sum that is invariant under Pachner moves. *TR and M. Vojinović*, [arXiv: 2201.02572](#).
 - ↪ This is a generalization of the state sum based on the classical $2BF$ action with the underlying 2-group structure
F. Girelli, H. Pfeiffer and E. M. Popescu, [arXiv:0708.3051](#).
- Is it TQFT?
 - ▶ Category theory essentials
 - ↪ Category $Hilb$, category $3Cob$, and category $3TCob$
 - ↪ Atiyah TQFT's axioms as a functorial conditions $Z : 3Cob \rightarrow Hilb$
 - ▶ Functor $Z : 3TCob \rightarrow Hilb$
 - ↪ Construction of the functor Z based on a 3-group, *TR and M. Vojinović*, [arXiv: 2412.21032](#).
 - ↪ Sketch of the proof that the functor satisfies the axioms, [arXiv: 2412.21032](#).
 - ↪ Lift from $Z : 3TCob \rightarrow Hilb$ to $\tilde{Z} : 3Cob \rightarrow Hilb$, [arXiv: 2412.21032](#).
- Conclusions

PRELIMINARIES

→ Theories of quantum gravity within the covariant approach are defined by quantizing the BF theory with constraints for the Lie group G ,

$$S_{BF} = \int_{\mathcal{M}_4} \langle B \wedge F \rangle_{\mathfrak{g}},$$

→ In order to overcome some issues of matter coupling in BF models of quantum gravity, a new approach is developed within the **framework of category theory**, based on a categorical generalization of the BF action – the so-called $2BF$ action ($BFCG$ action).

$$S_{2BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge G \rangle_{\mathfrak{h}}.$$

→ **Spin-cube model** of $4D$ gravity for the Poincaré 2-group.

A. Miković and M. Vojinović, 2012.

→ This result has opened the possibility of coupling matter with gravity in a linear fashion.

$$\text{2-crossed module } (L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{_, _\}_P)$$

- Groups G , H , and L ;
- Mappings ∂ and δ ($\partial\delta = 1_G$);
- Action $\triangleright$ of group G on all three groups;
- Mapping $\{_, _\}_P$ – *Paiffer lifting*:

$$\{_, _\}_P : H \times H \rightarrow L.$$

These groups and mappings must satisfy certain axioms in order to form a 2-crossed module:

1. $\delta(\{h_1, h_2\}_P) = \langle h_1, h_2 \rangle_P, \quad \forall h_1, h_2 \in H,$
2. $[l_1, l_2] = \{\delta(l_1), \delta(l_2)\}_P, \quad \forall l_1, l_2 \in L.$ Notation is $[l, k] = lkl^{-1}k^{-1};$
3. $\{h_1h_2, h_3\}_P = \{h_1, h_2h_3h_2^{-1}\}_P\partial(h_1) \triangleright \{h_2, h_3\}_P, \quad \forall h_1, h_2, h_3 \in H;$
4. $\{h_1, h_2h_3\}_P = \{h_1, h_2\}_P\{h_1, h_3\}_P\{\langle h_1, h_3 \rangle_P^{-1}, \partial(h_1) \triangleright h_2\}_P, \quad \forall h_1, h_2, h_3 \in H;$
5. $\{\delta(l), h\}_P\{h, \delta(l)\}_P = l(\partial(h) \triangleright l^{-1}), \quad \forall h \in H, \quad \forall l \in L.$

2-crossed module allows us to describe 3-gauge theory. J. F. Martins and R. Picken, 2011.; W. Wang 2014.

→ The structure of 2-crossed module leads to 3-connections, ordered triples (α, β, γ) , where α , β , and γ are differential form elements of algebras,

$$\begin{aligned}\alpha &= \alpha^\alpha{}_\mu \tau_\alpha dx^\mu, & \alpha &\in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}), \\ \beta &= \beta^a{}_{\mu\nu} t_a dx^\mu \wedge dx^\nu, & \beta &\in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}), \\ \gamma &= \gamma^A{}_{\mu\nu\rho} T_A dx^\mu \wedge dx^\nu \wedge dx^\rho, & \gamma &\in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}).\end{aligned}$$

→ Then we define line, surface, and volume holonomies,

$$g = \exp \int_\gamma \alpha, \quad h = \exp \int_S \beta, \quad l = \exp \int_V \gamma.$$

→ The corresponding fake 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$ is defined as:

$$\begin{aligned}\mathcal{F} &= d\alpha + \alpha \wedge \alpha - \partial\beta, & \mathcal{G} &= d\beta + \alpha \wedge^\triangleright \beta - \delta\gamma, \\ \mathcal{H} &= d\gamma + \alpha \wedge^\triangleright \gamma + \{\beta \wedge \beta\}_{\text{pf}}.\end{aligned}$$

→ For the manifold $\mathcal{M}_4$ and the 2-crossed module $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{-, -\}_{\text{pf}})$, or equivalently for the 3-curvature $(\mathcal{F}, \mathcal{G}, \mathcal{H})$, the 3BF action is defined as:

$$S_{3BF} = \int_{\mathcal{M}_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}}.$$

→ By adding constraints to the topological action, physically relevant models are defined:

2BF actions with constraints for:

- ▶ *Yang-Mills field*,
- ▶ and *Einstein-Cartan gravity*,

and 3BF actions with constraints describing

- ▶ *Klein-Gordon field*,
- ▶ *Dirac field*,
- ▶ *Weyl field*,
- ▶ and *Majorana field*.

coupled to gravity in the standard manner.

QUANTIZATION OF TOPOLOGICAL $3BF$ THEORY

Construction of the topological $3BF$ state sum based on the S_{3BF} action using the standard spinfoam quantization procedure.

$$Z = \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \mathcal{D}B \mathcal{D}C \mathcal{D}D \exp \left(i \int_{M_4} \langle B \wedge \mathcal{F} \rangle_{\mathfrak{g}} + \langle C \wedge \mathcal{G} \rangle_{\mathfrak{h}} + \langle D \wedge \mathcal{H} \rangle_{\mathfrak{l}} \right).$$

→ By formally integrating over the Lagrange multipliers B, C , and D , we obtain:

$$Z = \mathcal{N} \int \mathcal{D}\alpha \mathcal{D}\beta \mathcal{D}\gamma \delta(\mathcal{F})\delta(\mathcal{G})\delta(\mathcal{H}).$$

→ Discretization of the 3-connection:

- ▶ $\alpha \in \mathcal{A}^1(\mathcal{M}_4, \mathfrak{g}) \mapsto g_{\epsilon} \in G$ colors the edges $\epsilon = (jk) \in \Lambda_1$,
- ▶ $\beta \in \mathcal{A}^2(\mathcal{M}_4, \mathfrak{h}) \mapsto h_{\Delta} \in H$ colors the triangles $\Delta = (jkl) \in \Lambda_2$,
- ▶ $\gamma \in \mathcal{A}^3(\mathcal{M}_4, \mathfrak{l}) \mapsto l_{\tau} \in L$ colors the tetrahedra $\tau = (jk\ell m) \in \Lambda_3$.

$$\left. \begin{array}{ll} \int \mathcal{D}\alpha & \mapsto \prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \\ \int \mathcal{D}\beta & \mapsto \prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \\ \int \mathcal{D}\gamma & \mapsto \prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} \end{array} \right\} \longrightarrow \text{Discretization of the integral measure.}$$

QUANTIZATION OF TOPOLOGICAL $3BF$ THEORY

→ The condition $\delta(\mathcal{F})$ is discretized as

Girelli, Pfeiffer, and Popescu [arXiv: 0708.3051](#).

$$\delta(\mathcal{F}) = \prod_{(j k \ell) \in \Lambda_2} \delta_G(g_{j k \ell}), \quad \delta_G(g_{j k \ell}) = \delta_G(\partial(h_{j k \ell}) g_{k \ell} g_{j k} g_{j \ell}^{-1}).$$

→ The condition $\delta(\mathcal{G})$ is discretized as

TR and M. Vojinović, [arXiv: 2201.02572](#).

$$\delta(\mathcal{G}) = \prod_{(j k \ell m) \in \Lambda_3} \delta_H(h_{j k \ell m}),$$

$$\delta_H(h_{j k \ell m}) = \delta_H(\delta(l_{j k \ell m}) h_{j \ell m} (g_{\ell m} \triangleright h_{j k \ell}) h_{k \ell m}^{-1} h_{j k m}^{-1}).$$

→ The condition $\delta(\mathcal{H})$ is discretized as

TR and M. Vojinović, [arXiv: 2201.02572](#).

$$\delta(\mathcal{H}) = \prod_{(j k \ell m n) \in \Lambda_4} \delta_L(l_{j k \ell m n}),$$

$$\delta_L(l_{j k \ell m n}) = \delta_L(l_{j \ell m n}^{-1} h_{j \ell n} \triangleright' \{h_{\ell m n}, (g_{m n} g_{\ell m}) \triangleright h_{j k \ell}\} \triangleright l_{j k \ell n}^{-1} (h_{j k n} \triangleright' l_{k \ell m n}) l_{j k m n} h_{j m n} \triangleright' (g_{m n} \triangleright l_{j k \ell m n})).$$

...we obtain →

$$Z = \mathcal{N} \prod_{(j k) \in \Lambda_1} \int_G dg_{j k} \prod_{(j k \ell) \in \Lambda_2} \int_H dh_{j k \ell} \prod_{(j k \ell m) \in \Lambda_3} \int_L dl_{j k \ell m} \left(\prod_{(j k \ell) \in \Lambda_2} \delta_G(g_{j k \ell}) \right) \left(\prod_{(j k \ell m) \in \Lambda_3} \delta_H(h_{j k \ell m}) \right) \left(\prod_{(j k \ell m n) \in \Lambda_4} \delta_L(l_{j k \ell m n}) \right).$$

This expression becomes independent of the manifold triangulation by appropriate choice of the factor $\mathcal{N}$.

QUANTIZATION OF TOPOLOGICAL $3BF$ THEORY

Let $\mathcal{M}_d$ be a compact oriented combinatorial d -manifold, $d = 4$, and let $(L \xrightarrow{\delta} H \xrightarrow{\partial} G, \triangleright, \{-, -\}_{\text{pf}})$ be a 2-crossed module. The state sum of the topological 3-gauge theory is defined by the following expression:

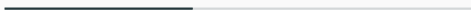
$$\begin{aligned} Z = & |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} |H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|} |L|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|+|\Lambda_3|-|\Lambda_4|} \\ & \times \left(\prod_{(jk) \in \Lambda_1} \int_G dg_{jk} \right) \left(\prod_{(jkl) \in \Lambda_2} \int_H dh_{jkl} \right) \left(\prod_{(jk\ell m) \in \Lambda_3} \int_L dl_{jk\ell m} \right) \\ & \times \left(\prod_{(jkl) \in \Lambda_2} \delta_G(g_{jkl}) \right) \left(\prod_{(jklm) \in \Lambda_3} \delta_H(h_{jklm}) \right) \left(\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}) \right). \end{aligned}$$

Where $|\Lambda_0|$ denotes the number of vertices, $|\Lambda_1|$ the number of edges, $|\Lambda_2|$ triangles, $|\Lambda_3|$ tetrahedra, and $|\Lambda_4|$ the number of 4-simplices in the triangulation.

↪ TR and M. Vojinović, [arXiv: 2201.02572](https://arxiv.org/abs/2201.02572).

↪ Behavior under Pachner moves has been analyzed and it is obtained that the state sum is invariant.

IS IT TQFT?



EXAMPLES OF A DAGGER SYMMETRIC MONOIDAL CATEGORY WITH A DUAL

The general dagger symmetric monoidal category with a dual and three relevant examples:

↪ *Hilb*,

↪ *3Cob*,

↪ *3TCob*.

	objects	morphisms	composition	identity morphism	tensor product	unit object	dagger	dual
$\mathcal{C}$	$A, \dots$	$f, g, \dots$	$\circ$	id_A	$\boxtimes$	1	$\dagger$	$*$
<i>Hilb</i>	$\mathcal{H}_A, \dots$	$\hat{M}, \hat{N}, \dots$	$\cdot$	$\hat{I}_{\mathcal{H}}$	$\otimes$	$\mathbb{C}$	$\dagger$	$*$
<i>3Cob</i>	$\Sigma_1, \dots$	$\mathcal{M}_A, \dots$	$\#$	id_{Σ}	$\sqcup$	$\emptyset$	$\dagger$	$\bar{\Sigma}$
<i>3TCob</i>	$\mathcal{T}(\Sigma_1), \dots$	$\mathcal{M}_A, \dots$	$\#$	id_{Σ}	$\sqcup$	$\emptyset$	$\dagger$	$\mathcal{T}(\bar{\Sigma})$

Example (Category *Hilb*)

The category *Hilb* of finite-dimensional Hilbert vector spaces is a dagger symmetric monoidal category with a dual, given as follows.

- The objects are finite-dimensional Hilbert spaces $\mathcal{H} \in \text{Obj}_{\text{Hilb}}$, and the morphisms $\hat{M} \in \text{Mor}_{\text{Hilb}}$ are linear operators between Hilbert spaces $\hat{M} : \mathcal{H}_A \rightarrow \mathcal{H}_B$.
- The source and target maps of $\hat{M} : \mathcal{H}_A \rightarrow \mathcal{H}_B$ are the domain and the codomain of the operator $s(\hat{M}) = \mathcal{H}_A$, $t(\hat{M}) = \mathcal{H}_B$.
- The composition of morphisms $\hat{M}, \hat{N} \in \text{Mor}_{\text{Hilb}}$ is the composition of operators $\hat{M} \cdot \hat{N}$.
- For each object, the identity morphism is the identity operator $\hat{I}_{\mathcal{H}}$ over that Hilbert space.
- The tensor product $\boxtimes$ is the ordinary tensor product $\otimes$ of linear algebra, the tensor product of vector spaces $\mathcal{H}_A \otimes \mathcal{H}_B$ for objects, and the tensor product of operators $\hat{M} \otimes \hat{N}$ for morphisms.
- The associator is trivial $(\mathcal{H}_A \otimes \mathcal{H}_B) \otimes \mathcal{H}_C = \mathcal{H}_A \otimes (\mathcal{H}_B \otimes \mathcal{H}_C)$.
- The unit object is the set of complex numbers $\mathbb{C}$.
- The left and right unitor maps are trivial $\mathbb{C} \otimes \mathcal{H} = \mathcal{H}$, $\mathcal{H} \otimes \mathbb{C} = \mathcal{H}$.
- The swap map is defined as $\sigma : \mathcal{H}_A \otimes \mathcal{H}_B \rightarrow \mathcal{H}_B \otimes \mathcal{H}_A$.
- The dagger is the Hermitian adjoint of the operator $\hat{M} : \mathcal{H}_A \rightarrow \mathcal{H}_B$, denoted $\dagger : \hat{M} \rightarrow \hat{M}^\dagger$, where the Hermitian adjoint $\hat{M}^\dagger : \mathcal{H}_B \rightarrow \mathcal{H}_A$ is the unique linear map that satisfies the property $\langle \phi | \hat{M} \psi \rangle = \langle \hat{M}^\dagger \phi | \psi \rangle$ for all vectors $|\phi\rangle \in \mathcal{H}_B$ and $|\psi\rangle \in \mathcal{H}_A$. The dagger acts trivially on objects, $\mathcal{H}^\dagger = \mathcal{H}$.
- The dual is just the dual Hilbert space $\mathcal{H}^*$.
- The unit and counit maps $\eta_{\mathcal{H}} : \mathbb{C} \rightarrow \mathcal{H}^* \otimes \mathcal{H}$ and $\epsilon_{\mathcal{H}} : \mathcal{H} \otimes \mathcal{H}^* \rightarrow \mathbb{C}$ are

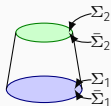
$$z \xrightarrow{\eta_{\mathcal{H}}} z \sum_i |i\rangle \otimes |i\rangle, \quad |i\rangle \otimes |j\rangle \xrightarrow{\epsilon_{\mathcal{H}}} \langle j|i\rangle,$$

for any $z \in \mathbb{C}$ and any orthonormal basis $\{|i\rangle \mid i = 1, \dots, \dim \mathcal{H}\}$ of the Hilbert space $\mathcal{H}$.

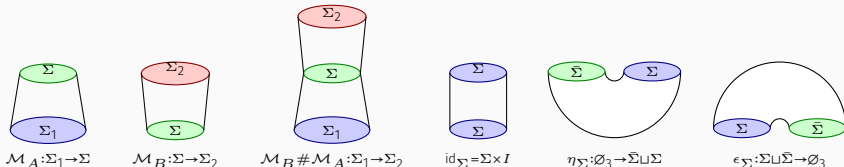
Example (Category $3Cob$)

The category $3Cob$ of oriented cobordisms between 3-dimensional manifolds is a dagger symmetric monoidal category with a dual, defined as follows.

- The objects $\Sigma \in Obj_{3Cob}$ are closed oriented 3-dimensional manifolds.
- The morphisms are compact oriented 4-dimensional manifolds $\mathcal{M}$.
- The source and target maps $\mathcal{M} : \Sigma_1 \rightarrow \Sigma_2$ are $s(\mathcal{M}) = \Sigma_1$, $t(\mathcal{M}) = \Sigma_2$.
- For each object Σ , there exists an identity morphism id_Σ given by a cylinder $[0, 1] \times \Sigma$, $\partial([0, 1] \times \Sigma) = \bar{\Sigma} \sqcup \Sigma$.
- The tensor product is a disjoint union of manifolds $\Sigma_1 \sqcup \Sigma_2$ and $\mathcal{M}_A \sqcup \mathcal{M}_B : \Sigma_1 \sqcup \Sigma_2 \rightarrow \Sigma_3 \sqcup \Sigma_4$.
- Trivial associator $(\Sigma_1 \sqcup \Sigma_2) \sqcup \Sigma_3 = \Sigma_1 \sqcup (\Sigma_2 \sqcup \Sigma_3)$.
- The unit object is the empty manifold $\emptyset_3$.
- The left and right unitor maps and the swap map are trivial $\emptyset_3 \sqcup \Sigma = \Sigma$, $\Sigma \sqcup \emptyset_3 = \Sigma$, $\Sigma_1 \sqcup \Sigma_2 = \Sigma_2 \sqcup \Sigma_1$.
- The dagger is a reversed cobordism $\mathcal{M}^\dagger : \Sigma_2 \rightarrow \Sigma_1$.
- The dual object is $\bar{\Sigma}$.
- The unit map and the counit map are $\eta_\Sigma : \emptyset_3 \rightarrow \bar{\Sigma} \sqcup \Sigma$, $\epsilon_\Sigma : \Sigma \sqcup \bar{\Sigma} \rightarrow \emptyset_3$.
- The composition of morphisms by the *connected sum* — *gluing* of manifolds along a common boundary.



$\mathcal{M} : \Sigma_1 \rightarrow \Sigma_2$ with the boundary $\partial\mathcal{M} = \bar{\Sigma}_1 \sqcup \Sigma_2$



Example (Category $3TCob$)

The category $3TCob$ of oriented cobordisms between triangulated 3-dimensional manifolds is a dagger symmetric monoidal category with a dual, defined as follows.

- The objects $\Sigma \in Obj_{3Cob}$ are closed oriented triangulated 3-dimensional manifolds.

- The morphisms Mor_{3TCob} are compact oriented 4-dimensional manifolds $\mathcal{M}$.

The triangulation of the bulk of the cobordism is not fixed, that is, the equivalence class of all possible triangulations of the same 4-dimensional manifold (that have the fixed triangulation on the boundary and that preserve the topology of the manifold) corresponds to a single morphism of the category.

- The source and target maps are defined as $s(\mathcal{M}) = \mathcal{T}(\Sigma_1)$, $t(\mathcal{M}) = \mathcal{T}(\Sigma_2)$.
- The composition of morphisms is given by the connected sum of manifolds, as in $3Cob$, but the difference is that **it is defined only if the triangulation of their common boundary matches**, for $\mathcal{M}_A : \mathcal{T}(\Sigma_1) \rightarrow \mathcal{T}(\Sigma)$ and $\mathcal{M}_B : \mathcal{T}'(\Sigma) \rightarrow \mathcal{T}(\Sigma_2)$, the composition is defined only when $\mathcal{T}(\Sigma) = \mathcal{T}'(\Sigma)$, and results in the cobordism $\mathcal{M}_B \# \mathcal{M}_A : \mathcal{T}(\Sigma_1) \rightarrow \mathcal{T}(\Sigma_2)$.

- For each object $\mathcal{T}(\Sigma)$, there exists an identity morphism $id_{\mathcal{T}(\Sigma)}$ given by a cylinder

$$[0, 1] \times \mathcal{T}(\Sigma), \quad \partial([0, 1] \times \mathcal{T}(\Sigma)) = \mathcal{T}(\bar{\Sigma}) \sqcup \mathcal{T}(\Sigma).$$

- The tensor product is a disjoint union of manifolds

$$\mathcal{T}(\Sigma_1) \sqcup \mathcal{T}(\Sigma_2)$$

$$\mathcal{M}_A \sqcup \mathcal{M}_B : \mathcal{T}(\Sigma_1) \sqcup \mathcal{T}(\Sigma_2) \rightarrow \mathcal{T}(\Sigma_3) \sqcup \mathcal{T}(\Sigma_4)$$

- As in the case of the $3Cob$ category, the associator, the left and the right unitor, and the swap map are all trivial.
- The unit object is the empty manifold $\emptyset_3$.
- The dagger is a reversed cobordism

$$\mathcal{M}^\dagger : \mathcal{T}(\Sigma_2) \rightarrow \mathcal{T}(\Sigma_1), \quad \partial \mathcal{M}^\dagger = \mathcal{T}(\bar{\Sigma}_2) \sqcup \mathcal{T}(\Sigma_1).$$

- The dual object is $\mathcal{T}(\bar{\Sigma})$, where the orientation reversal of a manifold is manifested in its triangulation by reversing the orientation of all links, triangles, and tetrahedrons.
- The unit map and the counit map are

$$\eta_{\mathcal{T}(\Sigma)} : \emptyset \rightarrow \mathcal{T}(\bar{\Sigma}) \sqcup \mathcal{T}(\Sigma), \quad \epsilon_{\mathcal{T}(\Sigma)} : \mathcal{T}(\Sigma) \sqcup \mathcal{T}(\bar{\Sigma}) \rightarrow \emptyset.$$

Correspondence between differential topology and quantum theory defined by a TQFT functor:

$$3Cob \xrightarrow{TQFT} Hilb.$$

Differential topology	Quantum theory
$3D$ manifold (space)	Hilbert space (states)
cobordism between 3-dimensional manifolds (spacetime)	operator (process)
composition of cobordisms	composition of operators
identity cobordism	identity operator
disjoint union of manifolds	tensor product
empty 3-dimensional manifold	1-dimensional Hilbert space
orientation reversal of the 4-dimensional manifold	Hermitian adjoint of an operator
orientation reversal of the 3-dimensional manifold	dual vector space

In our case, we define two functors:

$$3TCob \xrightarrow{Z} Hilb,$$

and

$$3Cob \xrightarrow{\tilde{Z}} Hilb.$$

Let us therefore spell out the axioms for Z . The axioms for $\tilde{Z}$ will be analogous.

Definition (**Functor** $Z : 3TCob \rightarrow Hilb$)

A functor $Z : 3TCob \rightarrow Hilb$ between two dagger symmetric monoidal categories with a dual $3TCob$ and $Hilb$:

1. *Objects*. Triangulated 3-dimensional manifolds $\mathcal{T}(\Sigma)$ are mapped to Hilbert space $\mathcal{H}$, $Z(\mathcal{T}(\Sigma)) = \mathcal{H}$.
2. *Morphisms*. 4-dimensional manifolds $\mathcal{M} : \mathcal{T}(\Sigma_1) \rightarrow \mathcal{T}(\Sigma_2)$ are mapped to operators $\hat{M} : \mathcal{H}_1 \rightarrow \mathcal{H}_2$, where $Z(\mathcal{T}(\Sigma_1)) = \mathcal{H}_1$ and $Z(\mathcal{T}(\Sigma_2)) = \mathcal{H}_2$, $Z(\mathcal{M}) = \hat{M}$.
3. *Composition of morphisms — functoriality*. The functor of a composition is a composition of functors

$$Z(\mathcal{M}_B \# \mathcal{M}_A) = \hat{M}_B \cdot \hat{M}_A.$$

4. *Identity morphism*. The functor Z maps the identity morphism of a triangulated manifold $\mathcal{T}(\Sigma)$ into the identity morphism of its corresponding Hilbert space $\mathcal{H}$,

$$Z(\text{id}_{\mathcal{T}(\Sigma)}) = \hat{I}_{\mathcal{H}}.$$

5. *Tensor product — multiplicativity*. The functor of a tensor product is a tensor product of functors

$$Z(\mathcal{T}(\Sigma_1) \sqcup \mathcal{T}(\Sigma_2)) = \mathcal{H}_1 \otimes \mathcal{H}_2, \quad Z(\mathcal{M}_A \sqcup \mathcal{M}_B) = \hat{M}_A \otimes \hat{M}_B,$$

6. *Unit object — normalization*. The functor maps the unit object into the unit object $Z(\emptyset_3) = \mathbb{C}$. And the identity morphism of the unit object in $TCob$, $\emptyset_4 = \text{id}_{\emptyset_3} : \emptyset_3 \rightarrow \emptyset_3$, maps to the identity morphism of the unit object in $\hat{I}_{\mathbb{C}} : \mathbb{C} \rightarrow \mathbb{C}$: $Z(\text{id}_{\emptyset_3}) = \hat{I}_{\mathbb{C}}$, $Z(\emptyset_4) = 1$.

7. *Symmetric monoidal maps*. The associator map, the swap map, and the left and right unitor maps are all trivial.

8. *Dagger — hermitivity*. The functor of a dagger is the dagger of a functor $Z(\mathcal{M}^\dagger) = \hat{M}^\dagger$.

9. *Dual — involution*. The functor of a dual is the dual of a functor $Z(\mathcal{T}(\bar{\Sigma})) = \mathcal{H}^*$.

10. *Unit and counit maps*. The unit map $\eta_{\mathcal{T}(\Sigma)} : \emptyset_3 \rightarrow \mathcal{T}(\bar{\Sigma}) \sqcup \mathcal{T}(\Sigma)$ and the counit map $\epsilon_{\mathcal{T}(\Sigma)} : \mathcal{T}(\Sigma) \sqcup \mathcal{T}(\bar{\Sigma}) \rightarrow \emptyset_3$ are mapped by the functor Z to their counterparts $\eta_{\mathcal{H}} : \mathbb{C} \rightarrow \mathcal{H}^* \otimes \mathcal{H}$ and $\epsilon_{\mathcal{H}} : \mathcal{H} \otimes \mathcal{H}^* \rightarrow \mathbb{C}$, $Z(\eta_{\mathcal{T}(\Sigma)}) = \eta_{\mathcal{H}}$, $Z(\epsilon_{\mathcal{T}(\Sigma)}) = \epsilon_{\mathcal{H}}$.

Preliminaries

↪ Given a 2-crossed module and a compact closed 4-dimensional manifold $\mathcal{M}$, we define a *state sum* (also called a *configuration integral* or *path integral*), as follows:

$$\begin{aligned} \mathcal{Z}_{\emptyset} = & |G|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|} |H|^{|\Lambda_0|-|\Lambda_1|+|\Lambda_2|-|\Lambda_3|} |L|^{-|\Lambda_0|+|\Lambda_1|-|\Lambda_2|+|\Lambda_3|-|\Lambda_4|} \\ & \times \left[\prod_{(j,k) \in \Lambda_1} \int_G dg_{jk} \right] \left[\prod_{(j,k,\ell) \in \Lambda_2} \int_H dh_{j k \ell} \right] \left[\prod_{(j,k,\ell,m) \in \Lambda_3} \int_L dl_{j k \ell m} \right] \\ & \times \left[\prod_{(j,k,\ell) \in \Lambda_2} \delta_G(g_{j k \ell}) \right] \left[\prod_{(j,k,\ell,m) \in \Lambda_3} \delta_H(h_{j k \ell m}) \right] \left[\prod_{(j,k,\ell,m,n) \in \Lambda_4} \delta_L(l_{j k \ell m n}) \right], \end{aligned}$$

↪ We now extend the definition of the statesum above to include manifolds with boundaries.

Preliminaries

→ Given a 4-dimensional manifold $\mathcal{M}$ with a possibly nontrivial boundary $\Sigma = \partial\mathcal{M}$, we introduce the state sum $\mathcal{Z}_\partial$ as follows:

$$\begin{aligned} \mathcal{Z}_\partial = & |G|^{-(|\Lambda_0| - \frac{|\Lambda_0, \Sigma|}{2}) + |\Lambda_1| - |\Lambda_2|} |H|^{(|\Lambda_0| - \frac{|\Lambda_0, \Sigma|}{2}) - (|\Lambda_1| - \frac{|\Lambda_1, \Sigma|}{2}) + |\Lambda_2| - |\Lambda_3|} \\ & |L|^{-(|\Lambda_0| - \frac{|\Lambda_0, \Sigma|}{2}) + (|\Lambda_1| - \frac{|\Lambda_1, \Sigma|}{2}) - (|\Lambda_2| - \frac{|\Lambda_2, \Sigma|}{2}) + |\Lambda_3| - |\Lambda_4|} \\ & \left[\prod_{(jk) \in \Lambda_1^*} \int_G dg_{jk} \right] \left[\prod_{(jkl) \in \Lambda_2^*} \int_H dh_{jkl} \right] \left[\prod_{(jk\ell m) \in \Lambda_3^*} \int_L dl_{jk\ell m} \right] \\ & \left[\prod_{(jk\ell) \in \Lambda_2^*} \delta_G(g_{jkl}) \right] \left[\prod_{(jklm) \in \Lambda_3^*} \delta_H(h_{jklm}) \right] \left[\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}) \right] \\ & \left[\prod_{(jkl) \in \Lambda_{2, \Sigma}} \delta_G(g_{jkl}) \right] \left[\prod_{(jklm) \in \Lambda_{3, \Sigma}} \delta_H(h_{jklm}) \right], \end{aligned}$$

where $\Lambda_{i, \Sigma}$ are the sets of simplices of dimension i on the boundary Σ (here $i \in \{0, 1, 2, 3\}$). The sets Λ_i^* contain simplices in the bulk only, namely $\Lambda_1^* = \Lambda_1 \setminus \Lambda_{1, \Sigma}$ is the set of edges of $\mathcal{T}(\mathcal{M})$ that are not the edges of $\mathcal{T}(\Sigma)$, and similarly for Λ_2^* and Λ_3^* .

When the boundary is empty, one can see the state sum $\mathcal{Z}(\mathcal{T}(\bar{\Sigma}_1), \mathcal{T}(\Sigma_2))$ reduces to state sum $\mathcal{Z}(\emptyset)$

$$\mathcal{Z}_\partial|_{\partial\mathcal{M}=\emptyset} = \mathcal{Z}_\emptyset.$$

Construction of the TQFT functor

We define a functor $Z : 3TCob \rightarrow Hilb$ in the following way.

- ↪ To each triangulated 3-dimensional manifold $\mathcal{T}(\Sigma) \in Obj_{3TCob}$ the functor Z assigns a Hilbert space $\mathcal{H}_\Sigma \in Obj_{Hilb}$. Denoting the numbers of edges, triangles and tetrahedra of the triangulation $\mathcal{T}(\Sigma)$ as $|\Lambda_{1,\Sigma}|$, $|\Lambda_{2,\Sigma}|$, and $|\Lambda_{3,\Sigma}|$ respectively, we choose the following Hilbert space of square-integrable functions over the three groups:

$$Z(\mathcal{T}(\Sigma)) = \mathcal{H} = \mathcal{L}^2(G^{|\Lambda_{1,\Sigma}|} \times H^{|\Lambda_{2,\Sigma}|} \times L^{|\Lambda_{3,\Sigma}|}).$$

such that

$$\dim \mathcal{L}^2(G) = |G|, \quad \dim \mathcal{L}^2(H) = |H|, \quad \dim \mathcal{L}^2(L) = |L|.$$

- ↪ Then, a basis element of the whole space $\mathcal{H}$ can be written as a tensor product of appropriate number of copies of the basis vectors from these three spaces,

$$|g, h, l\rangle \equiv \bigotimes_{i=1}^{|\Lambda_{1,\Sigma}|} |g_i\rangle \bigotimes_{j=1}^{|\Lambda_{2,\Sigma}|} |h_j\rangle \bigotimes_{k=1}^{|\Lambda_{3,\Sigma}|} |l_k\rangle \in \mathcal{H},$$

and the total dimension of $\mathcal{H}$ is given as:

$$\dim \mathcal{H} = |G|^{|\Lambda_{1,\Sigma}|} |H|^{|\Lambda_{2,\Sigma}|} |L|^{|\Lambda_{3,\Sigma}|}.$$

Construction of the TQFT functor

- ↪ To each 4-dimensional manifold $\mathcal{M}$ with the boundary $\partial\mathcal{M} = \mathcal{T}(\bar{\Sigma}_1) \sqcup \mathcal{T}(\Sigma_2)$, the functor Z assigns the operator $Z(\mathcal{M}) = \hat{M} : \mathcal{H}_1 \rightarrow \mathcal{H}_2$, where $\mathcal{H}_1 = Z(\mathcal{T}(\Sigma_1))$ and $\mathcal{H}_2 = Z(\mathcal{T}(\Sigma_2))$ as above. Then, the operator $\hat{M}$ is introduced as follows:

$$\hat{M} = \left[\prod_{(jk) \in \Lambda_1, \Sigma_1} \int_G dg'_{jk} \right] \left[\prod_{(jk\ell) \in \Lambda_2, \Sigma_1} \int_H dh'_{jk\ell} \right] \left[\prod_{(jk\ell m) \in \Lambda_3, \Sigma_1} \int_L dl'_{jk\ell m} \right] \\ \left[\prod_{(jk) \in \Lambda_1, \Sigma_2} \int_G dg_{jk} \right] \left[\prod_{(jk\ell) \in \Lambda_2, \Sigma_2} \int_H dh_{jk\ell} \right] \left[\prod_{(jk\ell m) \in \Lambda_3, \Sigma_2} \int_L dl_{jk\ell m} \right] \mathcal{Z}_{\partial} |g, h, l\rangle \otimes \langle g', h', l' | .$$

- ↪ Note that $\mathcal{Z}_{\partial} = \langle g, h, l | Z(\mathcal{M}) | g', h', l' \rangle$ is the kernel of the operator $\hat{M}$, while keeping in mind that $\mathcal{Z}_{\partial}$ is a function of the group elements living on the boundary.
- ↪ The definition of the kernel $\mathcal{Z}_{\partial}$ assumes that the whole 4-dimensional manifold $\mathcal{M}$ has been assigned some triangulation, $\mathcal{T}(\mathcal{M})$, while according to axiom 2, only the boundary $\partial\mathcal{M}$ is assigned a triangulation. Nevertheless, given the fact that $\mathcal{Z}_{\partial}$ is in fact invariant with respect to Pachner moves (as long as they do not modify the boundary), it has the same value for any choice of the triangulation in the bulk of $\mathcal{M}$. Therefore, the operator $\hat{M}$ does not in fact depend on the triangulation in the bulk, and for its explicit evaluation any choice $\mathcal{T}(\mathcal{M})$ will do, as long as it is compatible with the fixed triangulation of the boundary.

Construction of the TQFT functor

We now wish to remove the dependence on the triangulation of the 3-dimensional manifold, that is, to construct a functor

$$\tilde{Z} : 3Cob \rightarrow Hilb.$$

- ↪ To this end, the functor $Z : 3TCob \rightarrow Hilb$ introduced before serves as an intermediate step.
- ↪ The construction follows the key ideas by Yetter. (D. N. Yetter 1993, T. Porter 1996.)
- ↪ Different triangulations of the same 3-dimensional manifold form an equivalence class

$$Trng(\Sigma) = \{ \mathcal{T}(\Sigma) \mid \Sigma \in Obj_{3Cob} \}.$$

This set is infinite, since one can always refine any given triangulation without bound, but each set $Trng(\Sigma)$ contains at least one such distinguished “simplest” triangulation $\mathcal{T}_m(\Sigma) \in Trng(\Sigma)$.

- ↪ We construct the functor $\tilde{Z}$ as follows. Given any object $\Sigma \in Obj_{3Cob}$, find its minimal triangulation $\mathcal{T}_m(\Sigma)$, and then define the action of the functor $\tilde{Z}$ on Σ to be equal to the action of Z on $\mathcal{T}_m(\Sigma) \in Obj_{3TCob}$:

$$\tilde{Z}(\Sigma) = Z(\mathcal{T}_m(\Sigma)).$$

Next, given any cobordism $\tilde{\mathcal{M}} : \Sigma_1 \rightarrow \Sigma_2$ which is a morphism in category $3Cob$, one can find minimal triangulations of its source and target $\mathcal{T}_m(\Sigma_1)$ and $\mathcal{T}_m(\Sigma_2)$ respectively, and then define the action of the functor $\tilde{Z}$ on $\tilde{\mathcal{M}}$ to be equal to the action of Z on a corresponding morphism $\mathcal{M} : \mathcal{T}_m(\Sigma_1) \rightarrow \mathcal{T}_m(\Sigma_2)$ from the category $3TCob$:

$$\tilde{Z}(\tilde{\mathcal{M}}) = Z(\mathcal{M}).$$

Thus, we defined a map

$$3Cob \xrightarrow{\tilde{Z}} Hilb.$$

CONCLUSIONS

2BF topological state sum

- ↪ We construct the 2BF action for a general strict 2-group and any triangulation of any smooth d -dimensional spacetime manifold, $d \in \{3, 4\}$.
 - ↪ For $d = 3$, the constructed state sum is precisely the Jetter's model.
 - ↪ For $d = 4$, it coincides with Porter's TQFT for $d = 4$ and $n = 2$.
- ↪ 2BF state sum is a *topological invariant* of the manifold.
 - ↪ Girelli, Pfeiffer, Popescu, [arXiv: 0708.3051](#). Miković, Martins, [arXiv: 1006.0903](#).
- ↪ Representation theory for 2-groups (including the Poincare 2-group), has been developed in great detail. *Baez, Baratin, Freidel, Wise* [arXiv: 0812.4969](#).
- ↪ The topological invariant and TQFT for the Euclidean 2-group ($G = SO(4)$, $H = \mathbb{R}^4$) has also been studied in detail. *Asante, Dittrich, Girelli, Riello, Tsimiklis* [arXiv: 1908.05970](#).

2BF state sum

- ↪ For Poincare 2-group and 2BF action for GR, one possible quantization prescription leads to the *spincube model*. *A. Miković and M. Vojinović*, [arXiv: 1110.4694](#).

3BF topological state sum

- ↪ We formulate the 3BF state sum for the classical 3BF action in the case of a general semi-strict 3-group and 4-dimensional spacetime manifold.
 - ↪ It matches Porter's abstract definition of TQFT for $d = 4$ and $n = 3$.
- ↪ We find that it is a *topological invariant* of the manifold. *TR and M. Vojinović*, [arXiv: 2201.02572](#).

$3BF$ state sum as a TQFT

- ↪ Atiyah's axioms seamlessly translate into the criteria for a functor between the category of cobordisms and the category of vector space, $Z : 3Cob \rightarrow Hilb$
- ↪ As an intermediate step, we construct a functor $Z : 3TCob \rightarrow Hilb$, from which we obtain the functor $Z : 3Cob \rightarrow Hilb$, that is, a TQFT.
- ↪ These results tackle the Step 2 of the spinfoam quantization programme and provide the necessary mathematical foundations underlying the physically relevant quantum gravity model, namely the one based on the constrained $3BF$ action for gravity coupled to matter fields.
- ↪ However, to complete the **second step of the covariant spinfoam quantization procedure**, it is necessary to have **generalizations of the Peter-Weyl and Plancherel theorems** for the cases of 2-groups and 3-groups, mathematical results that are currently open problems.
- ↪ These theorems should **provide a decomposition of functions on a 3-group** into a sum over the corresponding irreducible representations of the 3-group. That would determine the **spectrum of labels of the simplices of the triangulation**, i.e., the range of values of fields living on the simplices of the triangulation, as was done in the case of the BF state sum.
- ↪ This result opens a way to **the third and final step of the covariant quantization procedure and the formulation of the quantum theory of gravity and matter of the Standard Model** by imposing appropriate constraints on the variables through modification of the amplitudes of the state sum.

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Thank you for your attention!

Definition (Category)

A category $\mathcal{C}$ consists of

- the objects of the category $\mathcal{C}$, denoted by $A, B, \dots \in \text{Obj}_{\mathcal{C}}$,
- and the morphisms between objects, denoted by $f, g, \dots \in \text{Mor}_{\mathcal{C}}$,

such that

- There exist source and target maps $s, t : \text{Mor}_{\mathcal{C}} \rightarrow \text{Obj}_{\mathcal{C}}$,

$$f : A \rightarrow B \quad s(f) = A, t(f) = B.$$

- The composition of morphisms

$$f : s(f) \rightarrow t(f), \quad g : s(g) \rightarrow t(g), \quad s(g) = t(f), \quad g \circ f : s(f) \rightarrow t(g),$$

$$\bullet \xleftarrow{g} \bullet \xleftarrow{f} \bullet = \bullet \xleftarrow{g \circ f} \bullet$$

- For each object $A \in \text{Obj}_{\mathcal{C}}$ of the category, there exists an identity morphism

$$\text{id}_A : A \rightarrow A.$$

Then, for each morphism $f : s(f) \rightarrow t(f)$, one has $\text{id}_{t(f)} \circ f = f, \quad f \circ \text{id}_{s(f)} = f.$

- For any three morphisms f, g , and h , when $s(g) = t(f)$ and $s(h) = t(g)$, composition is associative:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

$$\bullet \xleftarrow{h} \bullet \xleftarrow{g \circ f} \bullet = \bullet \xleftarrow{h \circ g} \bullet \xleftarrow{f} \bullet$$

CATEGORY THEORY ESSENTIALS

Definition (Symmetric monoidal category)

A symmetric monoidal category $(\mathcal{C}, \boxtimes, 1, \alpha, \lambda, \rho, \sigma)$ is a category $\mathcal{C}$ with objects $Obj_{\mathcal{C}}$ and morphisms $Mor_{\mathcal{C}}$, equipped with the following.

- The tensor product functor $\boxtimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$,

$$A, B \in Obj_{\mathcal{C}}, \quad A \boxtimes B,$$

and to each pair of morphisms $f, g \in Mor_{\mathcal{C}}$, such that $f : A_1 \rightarrow B_1$ and $g : A_2 \rightarrow B_2$, a morphism

$$f \boxtimes g : A_1 \boxtimes A_2 \rightarrow B_1 \boxtimes B_2.$$

- The associator map,

$$\alpha_{A,B,C} : (A \boxtimes B) \boxtimes C \rightarrow A \boxtimes (B \boxtimes C),$$

$$\begin{array}{ccccc} ((A \boxtimes B) \boxtimes C) \boxtimes D & \xrightarrow{\alpha_{A \boxtimes B, C, D}} & (A \boxtimes B) \boxtimes (C \boxtimes D) & \xrightarrow{\alpha_{A, B, C \boxtimes D}} & A \boxtimes (B \boxtimes (C \boxtimes D)) \\ \alpha_{A, B, C} \boxtimes id_D \searrow & & & & \nearrow id_A \boxtimes \alpha_{B, C, D} \\ (A \boxtimes (B \boxtimes C)) \boxtimes D & \xrightarrow{\alpha_{A, B \boxtimes C, D}} & A \boxtimes ((B \boxtimes C) \boxtimes D) & & \end{array}$$

- Unit object $1 \in Obj_{\mathcal{C}}$, and natural isomorphisms the left unitor map λ_A and the right unitor map ρ_A ,

$$\lambda_A : 1 \boxtimes A \rightarrow A \quad \text{and} \quad \rho_A : A \boxtimes 1 \rightarrow A,$$

$$\begin{array}{ccc} (A \boxtimes 1) \boxtimes B & \xrightarrow{\rho_A \boxtimes id_B} & A \boxtimes B \\ \alpha_{A, 1, B} \searrow & & \nearrow id_A \boxtimes \lambda_B \\ A \boxtimes (1 \boxtimes B) & & \end{array}$$

- The swap map $\sigma_{A,B} : A \boxtimes B \rightarrow B \boxtimes A$,

$$\begin{array}{ccc} A \boxtimes B & \xrightarrow{\sigma_{A,B}} & B \boxtimes A \\ f \boxtimes g \downarrow & & \downarrow g \boxtimes f \\ C \boxtimes D & \xrightarrow{\sigma_{C,D}} & D \boxtimes C. \end{array}$$

- α and σ map must satisfy the hexagon identity,

$$\begin{array}{ccccc} (A \boxtimes B) \boxtimes C & \xrightarrow{\alpha_{A,B,C}} & A \boxtimes (B \boxtimes C) & \xrightarrow{\sigma_{A, B \boxtimes C}} & (B \boxtimes C) \boxtimes A \\ \sigma_{A,B} \boxtimes id_C \downarrow & & & & \downarrow \alpha_{B,C,A} \\ (B \boxtimes A) \boxtimes C & \xrightarrow{\alpha_{B,A,C}} & B \boxtimes (A \boxtimes C) & \xrightarrow{id_B \boxtimes \sigma_{A,C}} & B \boxtimes (C \boxtimes A), \end{array}$$

and, σ , λ , and ρ must satisfy

$$\begin{array}{ccc} A \boxtimes 1 & \xrightarrow{\rho_{A,1}} & 1 \boxtimes A \\ \rho_A \searrow & & \nearrow \lambda_A \\ A & & \end{array}$$

- The swap map satisfies the inverse law

$$\sigma_{B,A} \circ \sigma_{A,B} = id_{A \boxtimes B}.$$

Definition (Dagger category)

A *dagger category* $(\mathcal{C}, \dagger)$ is a category equipped with a contravariant endofunctor, that reverses the directions of the morphisms, or equivalently $\dagger : \mathcal{C}^{op} \rightarrow \mathcal{C}$.

- For each morphism $f \in \text{Mor}_{\mathcal{C}}$, $f : A \rightarrow B$, the dagger functor gives the morphism $f^{\dagger} : B \rightarrow A$,

$$t(f^{\dagger}) = s(f), \quad s(f^{\dagger}) = t(f).$$

- For every object $A \in \text{Obj}_{\mathcal{C}}$,

$$(\text{id}_A)^{\dagger} = \text{id}_A.$$

- The dagger of the composition of two morphisms $f : A \rightarrow B$ and $g : B \rightarrow C$ is the composition of their daggers in reverse order,

$$(g \circ f)^{\dagger} = f^{\dagger} \circ g^{\dagger}.$$

- The dagger of the dagger of a morphism is the original morphism,

$$(f^{\dagger})^{\dagger} = f.$$

- The dagger functor acts as the identity for objects,

$$A^{\dagger} = A.$$

If the category is a dagger monoidal symmetric category, certain compatibility conditions hold between these structures of the category.

A *dagger symmetric monoidal category* $(\mathcal{C}, \boxtimes, 1, \alpha, \lambda, \rho, \sigma, \dagger)$ is a monoidal symmetric category equipped with a *dagger functor* $\dagger$ which is the identity for objects.

- For each pair of morphisms $f, g \in \text{Mor}_{\mathcal{C}}$, the dagger of a tensor product of morphisms f and g is the tensor product of daggers

$$(f \boxtimes g)^{\dagger} = f^{\dagger} \boxtimes g^{\dagger}.$$

Since the dagger functor acts trivially on objects, the analogous identity holds for the tensor product of objects.

- For any objects $A, B, C \in \text{Obj}_{\mathcal{C}}$, the dagger of the associator map is equal to the inverse of the associator,

$$\alpha_{A,B,C}^{\dagger} = \alpha_{A,B,C}^{-1} : A \boxtimes (B \boxtimes C) \rightarrow (A \boxtimes B) \boxtimes C.$$

- For each object $A \in \text{Obj}_{\mathcal{C}}$, the dagger of the left and right unitor maps are the inverses of those maps,

$$\lambda_A^{\dagger} = \lambda_A^{-1} : A \rightarrow 1 \boxtimes A \quad \text{and} \quad \rho_A^{\dagger} = \rho_A^{-1} : A \rightarrow A \boxtimes 1.$$

- For any two objects $A, B \in \text{Obj}_{\mathcal{C}}$, the dagger of the swap map is equal to the inverse of that map,

$$\sigma_{A,B}^{\dagger} = \sigma_{A,B}^{-1} : B \boxtimes A \rightarrow A \boxtimes B.$$

Definition (Symmetric monoidal category with a dual)

A symmetric monoidal category with a *dual* $(\mathcal{C}, \boxtimes, 1, \alpha, \lambda, \rho, \sigma, *)$ is a category where every object has its corresponding dual object.

- For every object $A \in \text{Obj}_{\mathcal{C}}$ of a monoidal category, there is a *dual object* A^* , the *unit map* and the *counit map*

$$\eta_A : 1 \rightarrow A^* \boxtimes A, \quad \epsilon_A : A \boxtimes A^* \rightarrow 1,$$

such that the following diagrams are the identity morphisms:

$$\begin{array}{ccc} A & \xrightarrow{\rho_A^{-1}} & A \boxtimes 1 \xrightarrow{\text{id}_A \boxtimes \eta_A} A \boxtimes (A^* \boxtimes A) \\ \lambda_A \uparrow & & \downarrow \alpha_{A, A^*, A}^{-1} \\ 1 \boxtimes A & \xleftarrow{\epsilon_A \boxtimes \text{id}_A} & (A \boxtimes A^*) \boxtimes A, \end{array}$$

$$\begin{array}{ccc} A^* & \xrightarrow{\lambda_{A^*}^{-1}} & 1 \boxtimes A^* \xrightarrow{\eta_A \boxtimes \text{id}_{A^*}} (A^* \boxtimes A) \boxtimes A^* \\ \rho_{A^*} \uparrow & & \downarrow \alpha_{A^*, A, A^*} \\ A^* \boxtimes 1 & \xleftarrow{\text{id}_{A^*} \boxtimes \epsilon_A} & A^* \boxtimes (A \boxtimes A^*). \end{array}$$

- The operation $*$ can be extended to a contravariant functor by mapping every morphism $f \in \text{Mor}_{\mathcal{C}}$ that maps $f : A \rightarrow B$ to a morphism $f^* : B^* \rightarrow A^*$.

A *dagger symmetric monoidal category with a dual* $(\mathcal{C}, \boxtimes, 1, \alpha, \lambda, \rho, \sigma, \dagger, *)$ is a dagger symmetric monoidal category and also a symmetric monoidal category with a dual, together with a condition to connect the dagger structure to the dual structure. The following is satisfied.

- For all $A \in \text{Obj}_{\mathcal{C}}$, the following diagram commutes:

$$\begin{array}{ccc}
 1 & \xrightarrow{\epsilon_A^\dagger} & A \boxtimes A^* \\
 \eta_A \searrow & & \swarrow \sigma_{A, A^*} \\
 & A^* \boxtimes A &
 \end{array}$$

- Since the dagger acts trivially on objects, for all objects $A \in \text{Obj}_{\mathcal{C}}$, one has the identity $(A^*)^\dagger = (A^\dagger)^*$.

Definition (Functor)

A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ between two dagger symmetric monoidal categories with a dual:

1. The functor F maps the objects and morphisms of one category to the objects and morphisms of another:

$$A \in \text{Obj}_{\mathcal{C}} \xrightarrow{F} F(A) \in \text{Obj}_{\mathcal{D}},$$

$$f : A \rightarrow B, f \in \text{Mor}_{\mathcal{C}} \xrightarrow{F} F(f) : F(A) \rightarrow F(B), F(f) \in \text{Mor}_{\mathcal{D}};$$

2. The source and target maps: $F(s(f)) = s(F(f)), \quad F(t(f)) = t(F(f));$
3. The functor F preserves the composition of morphisms, $F(g \circ f) = F(g) \circ F(f);$
4. The identity morphism: $F(\text{id}_A) = \text{id}_{F(A)};$
5. The tensor product $F(A \boxtimes_{\mathcal{C}} B) = F(A) \boxtimes_{\mathcal{D}} F(B), \quad F(f \boxtimes_{\mathcal{C}} g) = F(f) \boxtimes_{\mathcal{D}} F(g);$
6. The unit object and the identity morphism of the unit object $F(1_{\mathcal{C}}) = 1_{\mathcal{D}}, \quad F(\text{id}_{1_{\mathcal{C}}}) = \text{id}_{F(1_{\mathcal{C}})} = \text{id}_{1_{\mathcal{D}}};$
7. The associator map $\alpha_{A,B,C}$, the swap map $\sigma_{A,B}$, and the left and right unitor maps λ_A and ρ_A

$$F(\alpha_{A,B,C}) = \alpha_{F(A),F(B),F(C)}, \quad F(\sigma_{A,B}) = \sigma_{F(A),F(B)},$$

$$F(\lambda_A) = \lambda_{F(A)}, \quad F(\rho_A) = \rho_{F(A)};$$

8. The dagger map $F(f^{\dagger}) = F(f)^{\dagger};$
9. The dual satisfies the condition $F(A^*) = F(A)^*;$
10. The unit and counit map $F(\eta_A) = \eta_{F(A)}, \quad F(\epsilon_A) = \epsilon_{F(A)}.$

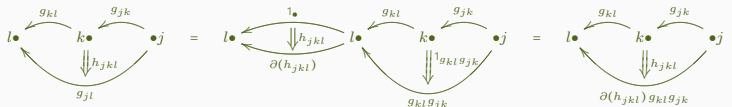
3-GAUGE THEORY ON TRIANGULATION

- Classical equations of motion impose the condition that the gauge connection is flat – that **every null-homotopic curve corresponds to the identity of the gauge group**.
- Within higher gauge theories, this condition is generalized by requiring that **the surface holonomy of the boundary 2-sphere of every 3-ball be trivial**.
- In the context of 3-gauge theory, the first condition remains unchanged, the second condition is generalized, while it is necessary to add the **condition of flatness of the boundary volume of the 4-simplex**.

Lemma 1

Girelli, Pfeiffer, and Popescu [arXiv: 0708.3051](https://arxiv.org/abs/0708.3051).

Consider the triangle (jkl) . The edges (jk) are labeled by group elements $g_{jk} \in G$ and the triangles (jkl) by elements $h_{jkl} \in H$.



The curvature $\gamma_1 = g_{kl} g_{jk}$ is the source, and the curvature $\gamma_2 = g_{jl}$ is the target of the surface 2-morphism $\Sigma : \gamma_1 \rightarrow \gamma_2$, labeled by the group element h_{jkl} ,

$$g_{jl} = \partial(h_{jkl}) g_{kl} g_{jk} .$$

Lemma 2

TR and M. Vojinović, arXiv: 2201.02572.

Consider the tetrahedron $(jklm)$. The tetrahedra $(jklm)$ are labeled by group elements $l_{jklm} \in L$.

$$\begin{aligned}
 & \text{Top diagram: } m \xrightarrow{g_{lm}} l \xrightarrow{g_{kl}} k \xrightarrow{g_{jk}} j \xrightarrow{g_{jm}} m. \text{ Faces: } h_{jlm}, h_{jkl}. \\
 & \quad = (g_{lm}g_{jl}, h_{jlm}) \#_2 (g_{lm} \#_1 (g_{kl}g_{jk}, h_{jkl})) = (g_{lm}g_{kl}g_{jk}, h_{jlm}(g_{lm} \triangleright h_{jkl})). \\
 & \text{Bottom diagram: } m \xrightarrow{g_{lm}} l \xrightarrow{g_{kl}} k \xrightarrow{g_{jk}} j \xrightarrow{g_{jm}} m. \text{ Faces: } h_{klm}, h_{jkm}. \\
 & \quad = (g_{km}g_{jk}, h_{jkm}) \#_2 ((g_{lm}g_{kl}, h_{klm}) \#_1 g_{jk}) = (g_{lm}g_{kl}g_{jk}, h_{jkm}h_{klm}).
 \end{aligned}$$

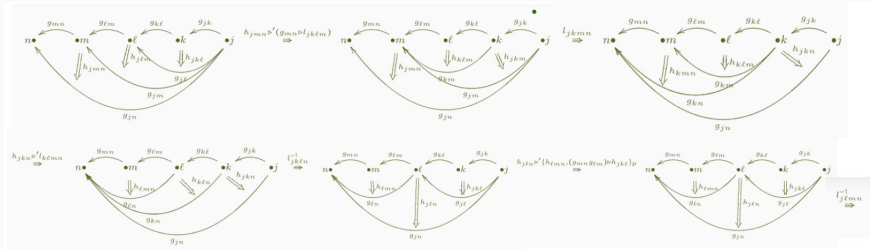
The mapping of the surface $\Sigma_1 : g_{lm}g_{kl}g_{jk} \rightarrow g_{jm}$ to the surface $\Sigma_2 : g_{lm}g_{kl}g_{jk} \rightarrow g_{jm}$ is determined by the element l_{jklm} :

$$h_{jkm}h_{klm} = \delta(l_{jklm})h_{jlm}(g_{lm} \triangleright h_{jkl}).$$

Lemma 3

TR and M. Vojinović, arXiv: 2201.02572.

We consider a 4-simplex, $(jklmn)$. We cut the 4-simplex volume along the surface $h_{jmn}g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl})$.



This brings us back to the initial surface!

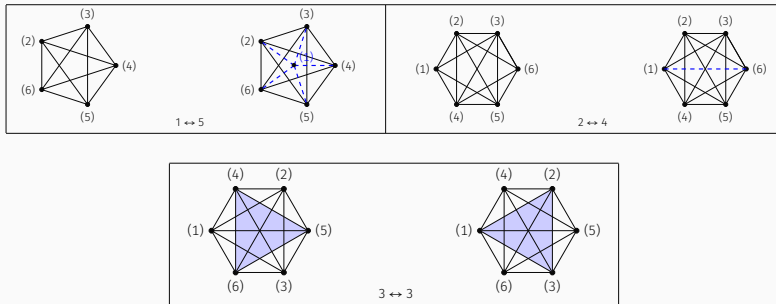
The obtained 3-morphism is the identity 3-morphism with source and target $\Sigma_1 = \Sigma_2 = h_{jmn}g_{mn} \triangleright (h_{jlm}g_{lm} \triangleright h_{jkl})$,

$$l_{jlmn}^{-1} h_{jln} \triangleright' \{h_{lmn}, (g_{mn}g_{lm}) \triangleright h_{jkl}\}_P l_{jklmn}^{-1} (h_{jkn} \triangleright' l_{klmn}) l_{jkmn} h_{jmn} \triangleright' (g_{mn} \triangleright l_{jklm}) = e.$$

PACHNER MOVES

- ↪ We analyzed the behavior of the constructed state sum under **Pachner moves**.
Pachner moves are local changes to triangulations that preserve topology, so any two triangulations of the same manifold are connected by a finite number of Pachner moves.
- ↪ In the $3D$ case, there are four Pachner moves — moves $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$ and their inverses, while in $4D$ there are five distinct Pachner moves — moves $3 \leftrightarrow 3$, $4 \leftrightarrow 2$, and $5 \leftrightarrow 1$ and their inverses.

Pachner moves in $4D$



SKETCH OF THE PROOF

Functor $Z : 3TCob \rightarrow Hilb$

1. Objects $Z(\mathcal{T}(\Sigma)) = \mathcal{H}$. ✓
2. Morphisms $Z(\mathcal{M}) = \hat{M}$. ✓
3. Composition of morphisms – functoriality $Z(\mathcal{M}_B \# \mathcal{M}_A) = \hat{M}_B \cdot \hat{M}_A$.
4. Identity morphism $Z(\text{id}_{\mathcal{T}(\Sigma)}) = \hat{I}_{\mathcal{H}}$.
5. Tensor product – multiplicativity $Z(\mathcal{T}(\Sigma_1) \sqcup \mathcal{T}(\Sigma_2)) = \mathcal{H}_1 \otimes \mathcal{H}_2$, $Z(\mathcal{M}_A \sqcup \mathcal{M}_B) = \hat{M}_A \otimes \hat{M}_B$.
6. Unit object – normalization $Z(\emptyset_3) = \mathbb{C}$. Identity morph. for the unit obj. $Z(\text{id}_{\emptyset_3}) = \hat{I}_{\mathbb{C}}$, $Z(\emptyset_4) = 1$.
7. Symmetric monoidal maps.
8. Dagger – hermitivity $Z(\mathcal{M}^\dagger) = \hat{M}^\dagger$.
9. Dual – involution $Z(\mathcal{T}(\bar{\Sigma})) = \mathcal{H}^*$.
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$$\hat{M}_{B\#A} = \hat{M}_B \cdot \hat{M}_A$$

Consider two 4-manifolds $\mathcal{M}_A : \mathcal{T}(\Sigma_A) \rightarrow \mathcal{T}(\Sigma)$ and $\mathcal{M}_B : \mathcal{T}(\Sigma) \rightarrow \mathcal{T}(\Sigma_B)$ with a common boundary $\mathcal{T}(\Sigma)$, the corresponding operators are:

$$\hat{M}_A = \left[\prod_{(jk) \in \Lambda_1, \Sigma_A} \int_G dg_{jk}^A \right] \left[\prod_{(jkl) \in \Lambda_2, \Sigma_A} \int_H dh_{jkl}^A \right] \left[\prod_{(jklm) \in \Lambda_3, \Sigma_A} \int_L dl_{jklm}^A \right] \\ \left[\prod_{(jk) \in \Lambda_1, \Sigma} \int_G dg_{jk} \right] \left[\prod_{(jkl) \in \Lambda_2, \Sigma} \int_H dh_{jkl} \right] \left[\prod_{(jklm) \in \Lambda_3, \Sigma} \int_L dl_{jklm} \right] \mathcal{Z}_{\partial}^A |g, h, l\rangle \otimes \langle g^A, h^A, l^A| ,$$

and similar for $\hat{M}_B$:

$$\hat{M}_B = \left[\prod_{(jk) \in \Lambda_1, \Sigma} \int_G dg'_{jk} \right] \left[\prod_{(jkl) \in \Lambda_2, \Sigma} \int_H dh'_{jkl} \right] \left[\prod_{(jklm) \in \Lambda_3, \Sigma} \int_L dl'_{jklm} \right] \\ \left[\prod_{(jk) \in \Lambda_1, \Sigma_B} \int_G dg_{jk}^B \right] \left[\prod_{(jkl) \in \Lambda_2, \Sigma_B} \int_H dh_{jkl}^B \right] \left[\prod_{(jklm) \in \Lambda_3, \Sigma_B} \int_L dl_{jklm}^B \right] \mathcal{Z}_{\partial}^B |g^B, h^B, l^B\rangle \otimes \langle g', h', l'| .$$

The composition $\hat{M}_B \cdot \hat{M}_A$ features two integrations over the simplices of $\mathcal{T}(\Sigma)$, namely over nonprimed and primed group elements, and a scalar product $\langle g', h', l' | g, h, l \rangle$. Let us look at these integrations for a single given edge in Λ_1, Σ , and perform the integration over the primed variable as follows:

$$\int_G dg \int_G dg' \mathcal{Z}_{\partial}^A(g, \dots) \mathcal{Z}_{\partial}^B(g', \dots) \langle g' | g \rangle = \int_G dg \frac{1}{|G|} \sum_{g' \in G} \mathcal{Z}_{\partial}^A(g, \dots) \mathcal{Z}_{\partial}^B(g', \dots) \chi_g(g') = \\ = \frac{1}{|G|} \int_G dg \mathcal{Z}_{\partial}^A(g, \dots) \mathcal{Z}_{\partial}^B(g, \dots) .$$

The condition reduces for the following condition for the state sums:

$$\mathcal{Z}_{\partial}^{B\#A} = \frac{1}{\dim \mathcal{H}} \left[\prod_{(jk) \in \Lambda_1, \Sigma} \int_G dg_{jk} \right] \left[\prod_{(jkl) \in \Lambda_2, \Sigma} \int_H dh_{jkl} \right] \left[\prod_{(jklm) \in \Lambda_3, \Sigma} \int_L dl_{jklm} \right] \mathcal{Z}_{\partial}^B \mathcal{Z}_{\partial}^A .$$

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$$Z(\text{id}_{\mathcal{T}(\Sigma)}) = \hat{I}_{\mathcal{H}}$$

For any triangulated manifold $\mathcal{T}(\Sigma) \in \text{Obj}_{3\text{T}Cob}$, the cylinder cobordism $\text{id}_{\mathcal{T}(\Sigma)} = [0, 1] \times \mathcal{T}(\Sigma)$ acts as its identity morphism in the category $3\text{T}Cob$. Given any cobordisms $\mathcal{M} : \mathcal{T}(\Sigma) \rightarrow \mathcal{T}(\Sigma')$ and $\mathcal{M}' : \mathcal{T}(\Sigma') \rightarrow \mathcal{T}(\Sigma)$ we have:

$$\text{id}_{\mathcal{T}(\Sigma)} \# \mathcal{M}' = \mathcal{M}', \quad \mathcal{M} \# \text{id}_{\mathcal{T}(\Sigma)} = \mathcal{M}.$$

one can apply the functor and employ the functoriality axiom, denoting $\hat{N} = Z(\text{id}_{\mathcal{T}(\Sigma)})$, to obtain

$$\hat{N} \cdot \hat{M}' = \hat{M}', \quad \hat{M} \cdot \hat{N} = \hat{M}.$$

Specifically, choosing $\mathcal{M}$ to be the cylinder cobordism itself, we obtain that $\hat{N}$ is a projector $\hat{N} \cdot \hat{N} = \hat{N}$. Then, assuming that the kernel of $\hat{N}$ is trivial in the space $\mathcal{H} = Z(\mathcal{T}(\Sigma))$, elementary linear algebra gives us

$$\hat{N} = \hat{I}_{\mathcal{H}}.$$

The identity operator $\hat{I}_{\mathcal{H}}$ can be written as

$$\begin{aligned} \hat{I}_{\mathcal{H}} &= \left[\prod_{(jk) \in \Lambda_{1,\Sigma}} \int_G dg'_{jk} \right] \left[\prod_{(jkl) \in \Lambda_{2,\Sigma}} \int_H dh'_{jkl} \right] \left[\prod_{(jklm) \in \Lambda_{3,\Sigma}} \int_L dl'_{jklm} \right] \\ &\quad \left[\prod_{(jk) \in \Lambda_{1,\Sigma}} \int_G dg_{jk} \right] \left[\prod_{(jkl) \in \Lambda_{2,\Sigma}} \int_H dh_{jkl} \right] \left[\prod_{(jklm) \in \Lambda_{3,\Sigma}} \int_L dl_{jklm} \right] \mathcal{Z}_{\partial}^{\text{id}} |g, h, l\rangle \otimes \langle g', h', l'|, \\ \mathcal{Z}_{\partial}^{\text{id}} &= (\dim \mathcal{H})^2 \prod_{(jk) \in \Lambda_{1,\Sigma}} \chi_{g_{jk}}(g'_{jk}) \prod_{(jkl) \in \Lambda_{2,\Sigma}} \chi_{h_{jkl}}(h'_{jkl}) \prod_{(jklm) \in \Lambda_{3,\Sigma}} \chi_{l_{jklm}}(l'_{jklm}). \end{aligned}$$

For finite groups G, H and L , upon substituting the kernel one obtains

$$\hat{I}_{\mathcal{H}} = \left[\prod_{(jk) \in \Lambda_{1,\Sigma}} \sum_{g_{jk} \in G} \right] \left[\prod_{(jkl) \in \Lambda_{2,\Sigma}} \sum_{h_{jkl} \in H} \right] \left[\prod_{(jklm) \in \Lambda_{3,\Sigma}} \sum_{l_{jklm} \in L} \right] |g, h, l\rangle \otimes \langle g, h, l|.$$

This has the form $\hat{I} = \sum_i |i\rangle \otimes \langle i|$ of the identity operator in an orth. basis $\{|i\rangle \mid i = 1, \dots, \dim \mathcal{H}\}$ of $\mathcal{H}$.

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$$Z(\mathcal{T}(\Sigma_1) \sqcup \mathcal{T}(\Sigma_2)) = \mathcal{H}_1 \otimes \mathcal{H}_2$$

Denoting $\Sigma = \Sigma_1 \sqcup \Sigma_2$, we can write the sets of simplices of the common refinement as a disjoint union of the sets of simplices of $\mathcal{T}(\Sigma_1)$ and $\mathcal{T}(\Sigma_2)$, and as a consequence, we have that

$$G^{|\Lambda_1, \Sigma|} = G^{|\Lambda_1, \Sigma_1|} \times G^{|\Lambda_1, \Sigma_2|}, \quad H^{|\Lambda_2, \Sigma|} = H^{|\Lambda_2, \Sigma_1|} \times H^{|\Lambda_2, \Sigma_2|}, \quad L^{|\Lambda_3, \Sigma|} = L^{|\Lambda_3, \Sigma_1|} \times L^{|\Lambda_3, \Sigma_2|}.$$

The space $\mathcal{H} = Z(\mathcal{T}(\Sigma))$ is given as

$$\begin{aligned} \mathcal{H} &= \mathcal{L}^2(G^{|\Lambda_1, \Sigma_1|} \times G^{|\Lambda_1, \Sigma_2|} \times H^{|\Lambda_2, \Sigma_1|} \times H^{|\Lambda_2, \Sigma_2|} \times L^{|\Lambda_3, \Sigma_1|} \times L^{|\Lambda_3, \Sigma_2|}) \\ &= \mathcal{L}^2(G^{|\Lambda_1, \Sigma_1|} \times H^{|\Lambda_2, \Sigma_1|} \times L^{|\Lambda_3, \Sigma_1|}) \otimes \mathcal{L}^2(G^{|\Lambda_1, \Sigma_2|} \times H^{|\Lambda_2, \Sigma_2|} \times L^{|\Lambda_3, \Sigma_2|}) = \mathcal{H}_1 \otimes \mathcal{H}_2. \end{aligned}$$

$$Z(\mathcal{M}_A \sqcup \mathcal{M}_B) = \hat{M}_A \otimes \hat{M}_B$$

Given the cobordisms and their corresponding operators

$$\mathcal{M}_A : \mathcal{T}(\Sigma_1) \rightarrow \mathcal{T}(\Sigma'_1), \quad \mathcal{M}_B : \mathcal{T}(\Sigma_2) \rightarrow \mathcal{T}(\Sigma'_2), \quad \mathcal{M}_A \sqcup \mathcal{M}_B : \mathcal{T}(\Sigma_1) \sqcup \mathcal{T}(\Sigma_2) \rightarrow \mathcal{T}(\Sigma'_1) \sqcup \mathcal{T}(\Sigma'_2),$$

$$\hat{M}_A = Z(\mathcal{M}_A), \quad \hat{M}_B = Z(\mathcal{M}_B), \quad \hat{M}_{A \sqcup B} = Z(\mathcal{M}_A \sqcup \mathcal{M}_B),$$

we can rewrite the equation into the following requirement for the kernels of the three operators:

$$\mathcal{Z}_{\partial}^{A \sqcup B} = \mathcal{Z}_{\partial}^A \mathcal{Z}_{\partial}^B.$$

We use

$$\begin{aligned} \prod_{(j k \ell) \in \Lambda_2, \mathcal{M}} \delta_G(g_{j k \ell}) &= \prod_{(j k \ell) \in \Lambda_2, \mathcal{M}_A} \delta_G(g_{j k \ell}) \prod_{(j k \ell) \in \Lambda_2, \mathcal{M}_B} \delta_G(g_{j k \ell}), \\ \prod_{(j k) \in \Lambda_{1, \mathcal{M}}^*} \int dg_{j k} &= \prod_{(j k) \in \Lambda_{1, \mathcal{M}_A}^*} \int dg_{j k} \prod_{(j k) \in \Lambda_{1, \mathcal{M}_B}^*} \int dg_{j k} \cdots \end{aligned}$$

And similarly over the groups H and L . Finally, it is straightforward to verify that the product of constant factors $\mathcal{N}_A$ and $\mathcal{N}_B$ gives the constant factor $\mathcal{N}_{\mathcal{M}}$.

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$$Z(\emptyset_3) = \mathbb{C}$$

One can see that the Hilbert space corresponding to an empty manifold $\emptyset_3$ is:

$$\mathcal{H} = Z(\emptyset_3) = \mathcal{L}^2(G^0 \times H^0 \times L^0) \cong \mathbb{C}.$$

In other words, when the boundary is empty, $\Sigma = \emptyset_3$, the Hilbert space $\mathcal{H}$ is just the set of complex numbers $\mathbb{C}$.

$$Z(\text{id}_{\emptyset_3}) = \hat{I}_{\mathbb{C}}$$

Using the fact that $\text{id}_{\emptyset_3} = \emptyset_4$, the l.h.s. is equal to $Z(\emptyset_4) = \mathcal{Z}_{\emptyset}^{\emptyset_4}$ where $\mathcal{Z}_{\emptyset}^{\emptyset_4}$ is the evaluation of $\mathcal{Z}$ over an empty manifold $\emptyset_4$. The r.h.s. becomes $\hat{I}_{\mathbb{C}} = 1$, thus obtaining

$$Z(\emptyset_4) = 1$$

Explicitly evaluating the state sum $\mathcal{Z}$ over an empty 4-manifold $\emptyset_4$, one indeed obtains

$$\mathcal{Z}_{\emptyset}^{\emptyset_4} = |G|^0 |H|^0 |L|^0 = 1.$$

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$$Z(\mathcal{M}^\dagger) = \hat{M}^\dagger$$

The dagger in the category $3TCob$ maps each 4-dimensional manifold $\mathcal{M} : \mathcal{T}(\Sigma_1) \rightarrow \mathcal{T}(\Sigma_2)$ to its reverse manifold $\mathcal{M}^\dagger : \mathcal{T}(\Sigma_2) \rightarrow \mathcal{T}(\Sigma_1)$, while the dagger in the category $Hilb$ maps each operator $\hat{M}$ to its Hermitian adjoint operator $\hat{M}^\dagger$. Then, the hermitian axiom requires that the functor Z commutes with the dagger, and one can rewrite this condition in terms of the state sums as: $\mathcal{Z}_\partial = \mathcal{Z}_\partial^\dagger$. The state sum $\mathcal{Z}_\partial$ corresponding to the manifold $\mathcal{M}$ is

$$\begin{aligned} \mathcal{Z}_\partial = & \mathcal{N}_\mathcal{M} \left[\prod_{(jk) \in \Lambda_1^*} \int_G dg_{jk} \right] \left[\prod_{(jkl) \in \Lambda_2^*} \int_H dh_{jkl} \right] \left[\prod_{(jk\ell m) \in \Lambda_3^*} \int_L dl_{jk\ell m} \right] \\ & \left[\prod_{(jkl) \in \Lambda_2} \delta_G(g_{jk}) \right] \left[\prod_{(jklm) \in \Lambda_3} \delta_H(h_{jklm}) \right] \left[\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(l_{jk\ell mn}) \right], \end{aligned}$$

while the state sum $\mathcal{Z}_\partial^\dagger$ corresponding to the reversed manifold $\mathcal{M}^\dagger$ is

$$\begin{aligned} \mathcal{Z}_\partial^\dagger = & \mathcal{N}_{\mathcal{M}^\dagger} \left[\prod_{(jk) \in \Lambda_1^*} \int_G dg_{jk} \right] \left[\prod_{(jkl) \in \Lambda_2^*} \int_H dh_{jkl} \right] \left[\prod_{(jk\ell m) \in \Lambda_3^*} \int_L dl_{jk\ell m} \right] \\ & \left[\prod_{(jkl) \in \Lambda_2} \delta_G(\overline{g_{jk}}) \right] \left[\prod_{(jklm) \in \Lambda_3} \delta_H(\overline{h_{jklm}}) \right] \left[\prod_{(jk\ell mn) \in \Lambda_4} \delta_L(\overline{l_{jk\ell mn}}) \right]. \end{aligned}$$

The orientation reversal does not change the number of simplices of a triangulation, so we have $\mathcal{N}_\mathcal{M} = \mathcal{N}_{\mathcal{M}^\dagger}$. However, the consequences of orientation reversal on the group elements are not trivial, so we prove that

$$\delta_G(g_{jk\ell}) = \delta_G(\overline{g_{jk\ell}}), \quad \delta_H(h_{jklm}) = \delta_H(\overline{h_{jklm}}), \quad \delta_L(l_{jk\ell mn}) = \delta_L(\overline{l_{jk\ell mn}}).$$

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$$\mathcal{H}_{\bar{\Sigma}} = \mathcal{H}_{\Sigma}^*$$

Let $\mathcal{H}_{\Sigma}$ denote the Hilbert space associated with the triangulated 3-dimensional manifold $\mathcal{T}(\Sigma)$, namely $\mathcal{H}_{\Sigma} = Z(\mathcal{T}(\Sigma))$, and let $\mathcal{H}_{\bar{\Sigma}}$ denote the Hilbert space associated with the triangulated manifold $\mathcal{T}(\bar{\Sigma})$, namely $\mathcal{H}_{\bar{\Sigma}} = Z(\mathcal{T}(\bar{\Sigma}))$, where $\mathcal{T}(\bar{\Sigma})$ is the triangulation obtained from the triangulation $\mathcal{T}(\Sigma)$ by reversing the orientation of all its links, triangles and tetrahedra. Then, the involutory axiom states that $\mathcal{H}_{\bar{\Sigma}} = \mathcal{H}_{\Sigma}^*$, where $\mathcal{H}_{\Sigma}^*$ denotes the dual Hilbert space of $\mathcal{H}_{\Sigma}$.

To demonstrate this, note that the triangulation $\mathcal{T}(\Sigma)$ and the triangulation $\mathcal{T}(\bar{\Sigma})$ have equal numbers of simplices, so their corresponding Hilbert spaces $\mathcal{H}_{\Sigma}$ and $\mathcal{H}_{\bar{\Sigma}}$ are isomorphic, and also both are isomorphic to $\mathcal{H}_{\Sigma}^*$,

$$\mathcal{H}_{\bar{\Sigma}} \cong \mathcal{H}_{\Sigma} \cong \mathcal{H}_{\Sigma}^* .$$

Moreover, both the orientation reversal operation and the operation of taking the dual of a Hilbert space are involutions,

$$\mathcal{T}(\bar{\bar{\Sigma}}) = \mathcal{T}(\Sigma) , \quad \mathcal{H}_{\Sigma}^{**} = \mathcal{H}_{\Sigma} .$$

Therefore, nothing prevents us from making an identification between orientation reversal and taking the dual, so we can basically define that

$$Z(\mathcal{T}(\bar{\Sigma})) = \mathcal{H}_{\Sigma}^* .$$

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$$Z(\eta_{\mathcal{T}(\Sigma)}) = \eta_{\mathcal{H}}$$

The cobordisms id_{Σ} , η_{Σ} and ϵ_{Σ} are formally different since they have different sources and targets

$$\text{id}_{\Sigma} : \mathcal{T}(\Sigma) \rightarrow \mathcal{T}(\Sigma), \quad \eta_{\Sigma} : \emptyset_3 \rightarrow \mathcal{T}(\bar{\Sigma}) \sqcup \mathcal{T}(\Sigma), \quad \epsilon_{\Sigma} : \mathcal{T}(\Sigma) \sqcup \mathcal{T}(\bar{\Sigma}) \rightarrow \emptyset_3.$$

They are all described by the same manifold $\mathcal{M}$ that has the same boundary $\partial\mathcal{M} = \mathcal{T}(\bar{\Sigma}) \sqcup \mathcal{T}(\Sigma)$, therefore

$$\mathcal{Z}_{\partial}^{\text{id}} = \mathcal{Z}_{\partial}^{\eta} = \mathcal{Z}_{\partial}^{\epsilon}.$$

The domains and codomains are:

$$Z(\text{id}_{\Sigma}) : \mathcal{H} \rightarrow \mathcal{H}, \quad Z(\eta_{\Sigma}) : \mathbb{C} \rightarrow \mathcal{H}^* \otimes \mathcal{H}, \quad Z(\epsilon_{\Sigma}) : \mathcal{H} \otimes \mathcal{H}^* \rightarrow \mathbb{C}.$$

The unit operator is written as:

$$Z(\eta_{\Sigma}) = \left[\prod_{(jk) \in \Lambda_{1,\Sigma}} \int_G dg'_{jk} \right] \left[\prod_{(jk\ell) \in \Lambda_{2,\Sigma}} \int_H dh'_{jk\ell} \right] \left[\prod_{(jk\ell m) \in \Lambda_{3,\Sigma}} \int_L dl'_{jk\ell m} \right] \\ \left[\prod_{(jk) \in \Lambda_{1,\Sigma}} \int_G dg_{jk} \right] \left[\prod_{(jk\ell) \in \Lambda_{2,\Sigma}} \int_H dh_{jk\ell} \right] \left[\prod_{(jk\ell m) \in \Lambda_{3,\Sigma}} \int_L dl_{jk\ell m} \right] \mathcal{Z}_{\partial}^{\eta} \langle g, h, l | \otimes | g', h', l' \rangle \otimes 1.$$

To see that it has precisely the form as the unit operator in Hilbert space, we can act with it on an arbitrary number $z \in \mathbb{C}$, and for the choice of finite groups G, H and L the integrals become normalized sums, and one obtains

$$Z(\eta_{\Sigma})z = z \left[\prod_{(jk) \in \Lambda_{1,\Sigma}} \sum_{g_{jk} \in G} \right] \left[\prod_{(jk\ell) \in \Lambda_{2,\Sigma}} \sum_{h_{jk\ell} \in H} \right] \left[\prod_{(jk\ell m) \in \Lambda_{3,\Sigma}} \sum_{l_{jk\ell m} \in L} \right] \langle g, h, l | \otimes | g, h, l \rangle,$$

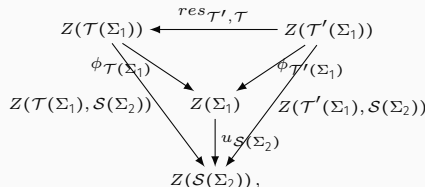
which has precisely the same form $\eta_{\mathcal{H}}(z) = z \sum_i \langle i | \otimes | i \rangle$ as the unit map of the Hilbert space.

$$Z(\epsilon_{\mathcal{T}(\Sigma)}) = \epsilon_{\mathcal{H}}$$

Similarly, we demonstrate the counit axiom.

YETTER'S TRIANGULATIONS AND TQFT'S

Using the definition of the colimit, we define the Hilbert space $Z(\Sigma_1)$ to be the colimit of the diagram $(Z(\mathcal{T}(\Sigma_1)), res_{\mathcal{T}', \mathcal{T}})$ and denote the family of morphisms from $Z(\mathcal{T}(\Sigma_1))$ to $Z(\Sigma_1)$ as $\phi_{\mathcal{T}(\Sigma_1)} : Z(\mathcal{T}(\Sigma_1)) \rightarrow Z(\Sigma_1)$, for every triangulation $\mathcal{T}$ of Σ_1 . If we keep the triangulation $\mathcal{S}(\Sigma_2)$ fixed, the maps $Z(\mathcal{T}(\Sigma_1), \mathcal{S}(\Sigma_2))$ form a cocone, and thus there is a unique map $u_{\mathcal{S}(\Sigma_2)} : Z(\Sigma_1) \rightarrow Z(\mathcal{S}(\Sigma_2))$, as shown on the diagram,



such that $u_{\mathcal{S}(\Sigma_2)} \phi_{\mathcal{T}(\Sigma_1)} = Z(\mathcal{T}(\Sigma_1), \mathcal{S}(\Sigma_2))$ for all triangulations $\mathcal{T}(\Sigma_1)$. Similarly, one can define $Z(\Sigma_2)$ as the colimit of $(Z(\mathcal{S}(\Sigma_2)), res_{\mathcal{S}', \mathcal{S}})$ and denote the maps from $Z(\mathcal{S}(\Sigma_2))$ to $Z(\Sigma_2)$ as $\phi_{\mathcal{S}(\Sigma_2)} : Z(\mathcal{S}(\Sigma_2)) \rightarrow Z(\Sigma_2)$ for each triangulation $\mathcal{S}$ of Σ_2 . From comutativity of the diagram it follows that $u_{\mathcal{S}(\Sigma_2)} = res_{\mathcal{S}', \mathcal{S}} u_{\mathcal{S}'(\Sigma_2)}$. One then obtains a map

$$Z(\Sigma_1, \Sigma_2) = \phi_{\mathcal{S}(\Sigma_2)} u_{\mathcal{S}(\Sigma_2)} : Z(\Sigma_1) \rightarrow Z(\Sigma_2),$$

for any triangulation $\mathcal{S}$ of Σ_2 , hence a TQFT.